Dealing with imbalanced classes

Logistic Regression case study

confusion matrix

		Predicted	
Actuals	0	TN	FP
	1	FN	TP

$$\text{Accuracy} = \frac{TP + TN}{TN + FP + FN + TP}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

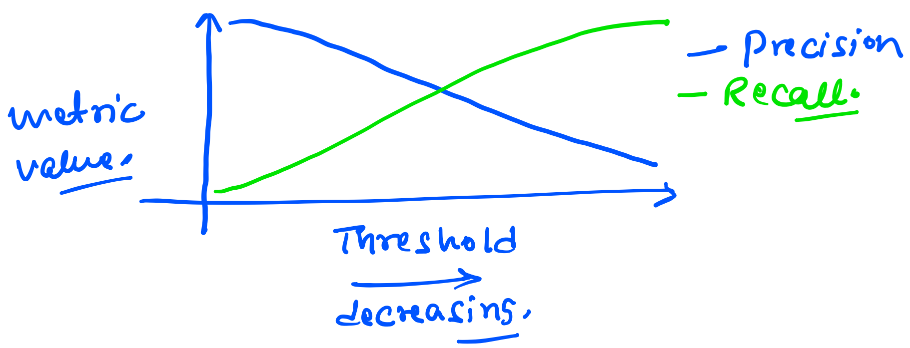
$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-score} = \frac{2PR}{P+R}$$

Threshold Tuning:

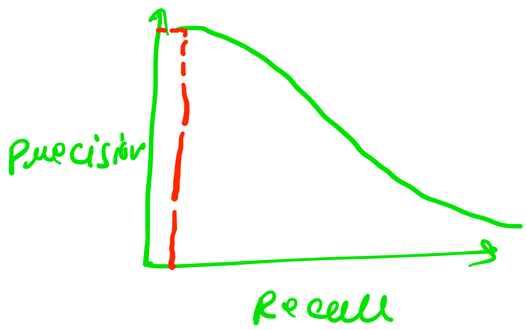
Lower → inc. in positives → inc. in false positives
recall increases.

Higher → Predict fewer positives → more false negatives
Precision higher.



Roc-Auc curve:

PR-AUC curve:



$$FPR = \frac{FP}{FP+TN}$$

$$TPR = \frac{TP}{TP+FN}$$

Handle imbalances:

1. Using correct metrics: Precision, Recall, F1-score, PR-Auc curve.

2. tune threshold

3. class weight logistic regression:

$$J(w) = -\frac{1}{n} \sum_{i=1}^{n} w_{y_i} \left[\left(y_i (\log \hat{y}_i \right) + (1-y_i) \log(1-\hat{y}_i) \right]$$

$$weight_c = \frac{n}{K \cdot n_c}$$

n = total no. of samples
K = total no. of classes
n_c = no. of samples for class c.

$n = 1000$

$K = 2$

$n_0 = 900, \quad n_1 = 100$

$$w_0 = \frac{1000}{2 \times 900} = \frac{10}{18} = 0.55$$

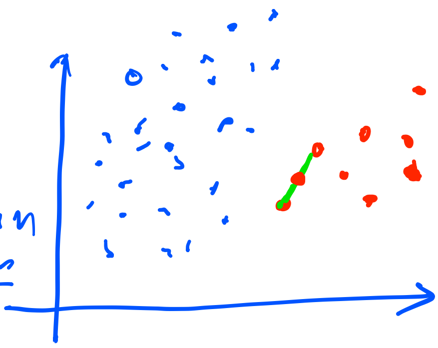
$$w_1 = \frac{1000}{2 \times 100} = \frac{10}{2} = 5.0$$

(4) Oversample the minority cases:

Duplicate the minority rows until numbers match.

(5) Undersampling the majority class:

(6) SMOTE:



(0) find a minority sample

(1) find neighbors → Euclidean distance.

(2) interpolate.

(3) Repeat until no. matches with majority

$$x_{new} = x_i + \lambda \cdot (x_{neighbor} - x_i), \quad 0 \leq \lambda \leq 1$$

$A = (10, 80), \quad B = (20, 100) \quad \lambda = 0.4$

$x_{new} = (14, 88)$

7. combine over and undersampling

900 : 100 → 300 : 300

8. stratified split :-

$$\frac{n_0}{n_1} \approx \frac{n_{0\_train}}{n_{1\_train}} \approx \frac{n_{0\_test}}{n_{1\_test}}$$

9. collect more minority data.